

A

Thread 1 — tumor-intrinsic (N = 518, 5 long-course nCRT cohorts)

DSB / HDR repair

Z = +3.17 ★★★★★
P = 0.002

Tumor cell-cycle

Z = +3.21 ★★★★★
P = 0.001

E2F / MYC

Z = +2.79 ★★★
P = 0.005

EMT

Z = +1.61
P = 0.106

Thread 2 — immune (N = 518, 5 cohorts · +Akiyoshi N = 816, 6 sources)

CD8-cytotoxic (5 cohorts)

Z = +2.00 ★★
P = 0.046

CD8-cytotoxic + Akiyoshi 2023 ★

Z = +3.29 ★★★★★
P = 0.001

T-cell infiltration

Z = +0.84
P = 0.399

B-cell infiltration

Z = +0.28
P = 0.781

Per-cohort Z:

● GSE35452 (46)

● GSE45404 (80)

● GSE56699 (44)

● GSE133057 (33)

● GSE87211 (353)

● GSE216616 Akiyoshi '23

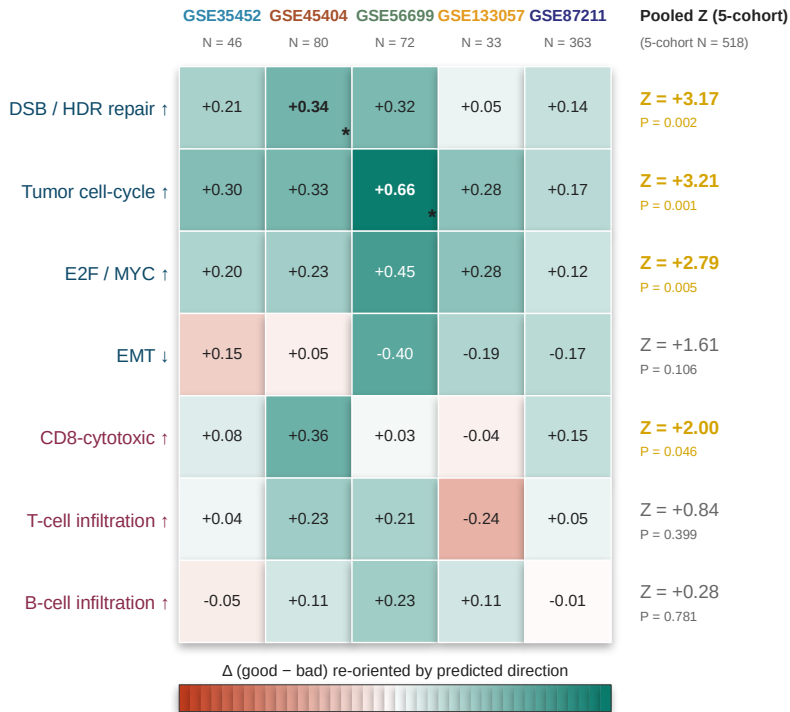
◆ pooled Stouffer Z (T1)

◆ pooled Z (T2)

◆ Akiyoshi-augmented

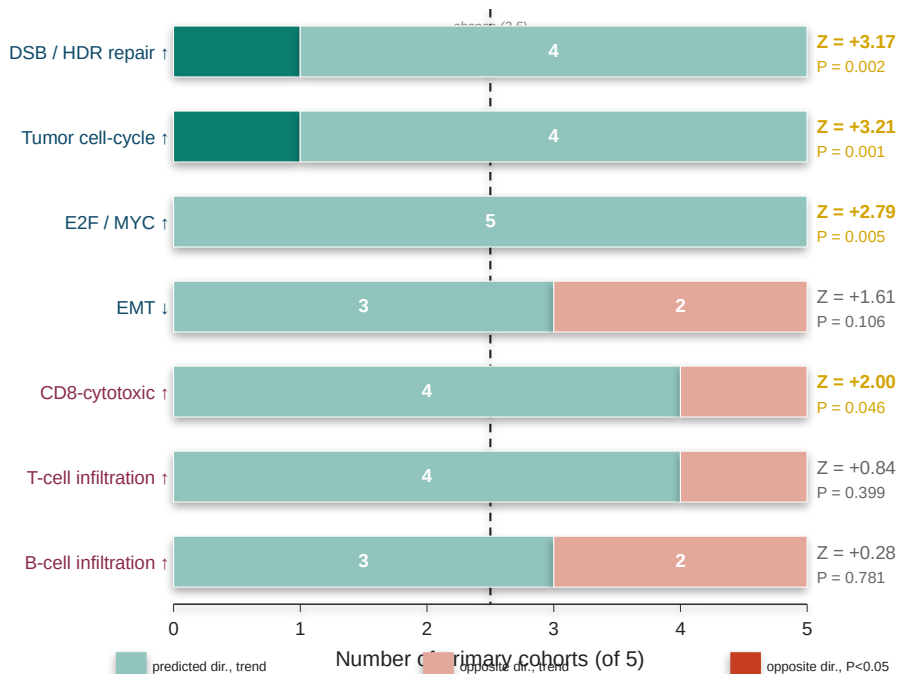
Stouffer pooled Z (predicted direction-oriented)

B

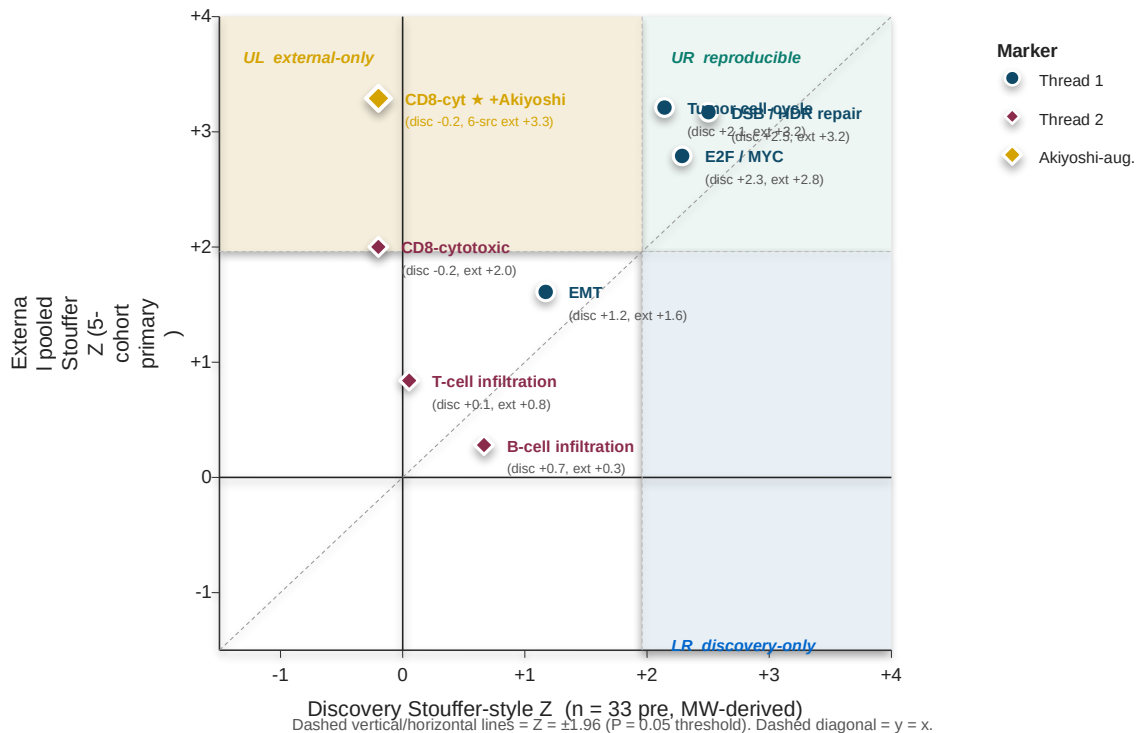


* = per-cohort Mann-Whitney $P < 0.05$. Pooled Z values gold-highlighted when $|Z| \geq 1.96$ ($P \leq 0.05$).

C



D

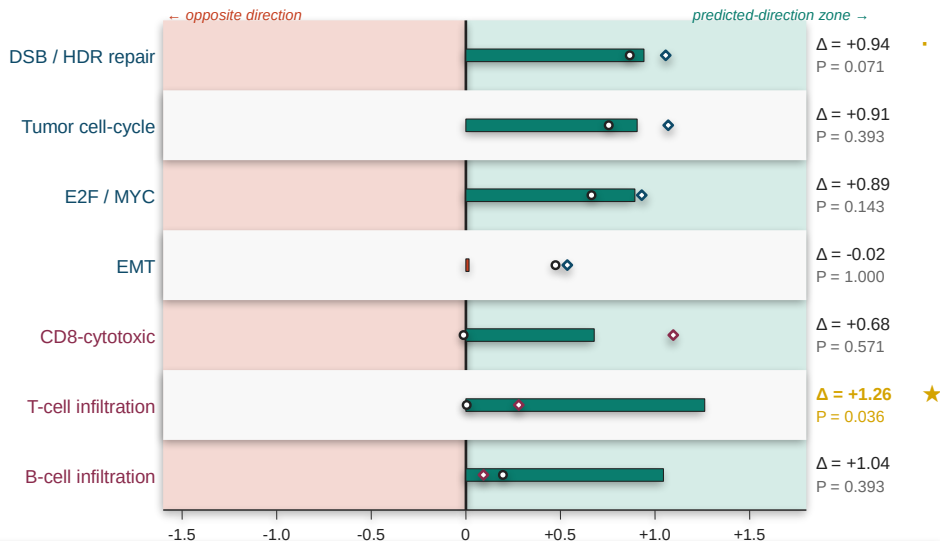


CD8-cytotoxic sits in the UL (external-only) quadrant in discovery (n=33 under-powered for immune), but Akiyoshi-augmented moves toward UR.

E

SC-RT-matched external validation — GSE254249 (Gao et al. Cancer Cell 2025)

$n = 8$ post-TNT bulk RNA-seq · SC-RT 5×5 Gy + FOLFOXIRI · 5 CR vs 3 non-CR



Three-regimen regimen-agnostic convergence

- Discovery SC-RT + FOLFOX/CAPOX (N=33 pre-RT): LASSO AUC 0.745 [0.56, 0.90]
- LC-CRT + concurrent capecitabine (N=518–816): T1 DSB Z=+3.17, cellcycle Z=+3.21, E2F/MYC Z=+2.79; T2 CD8 Z=+3.29
- **SC-RT + FOLFOXIRI (N=8 post-TNT): 7/7 signatures concordant — binomial sign $P = 0.016$ — Tcell_infil MW $P = 0.036$ ★**

Baseline predictor biology reproduces across both RT fractionation and chemo-timing/backbone axes.

Markers:

SC-RT Δ (bar)

Discovery Δ (circle)

LC-CRT pooled Z / 3 (diamond)